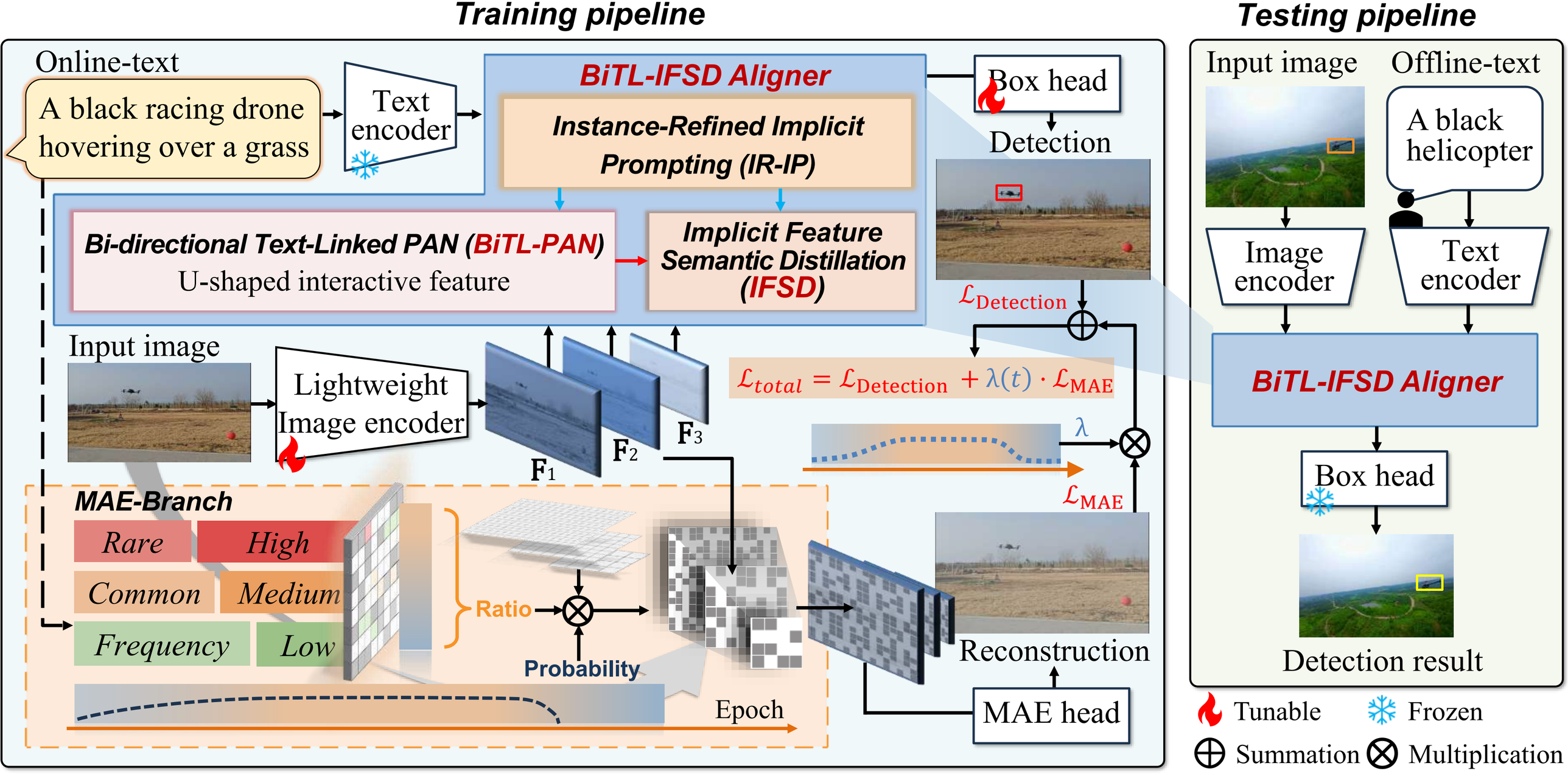
Training pipeline
Testing pipeline
Online-text
A black racing drone hovering over a grass
Text encoder
BiTL-IFSD Aligner
Instance-Refined Implicit Prompting (IR-IP)
Bi-directional Text-Linked PAN (BiTL-PAN)
U-shaped interactive feature
Implicit Feature Semantic Distillation (IFSD)
Box head
Detection
$\mathcal{L}_{\text{Detection}}$
$\mathcal{L}_{total} = \mathcal{L}_{\text{Detection}} + \lambda(t) \cdot \mathcal{L}_{\text{MAE}}$
Input image
Lightweight Image encoder
F_1
F_2
F_3
λ
$\mathcal{L}_{\text{MAE}}$
MAE-Branch
Rare
High
Common
Medium
Frequency
Low
Ratio
Probability
Epoch
Reconstruction
MAE head
Input image
Offline-text
A black helicopter
Image encoder
Text encoder
BiTL-IFSD Aligner
Box head
Detection result
Tunable
Frozen
Summation
Multiplication